6.1200 *Mathematics for Computer Science* 22 October 2026
Problem Set 9
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Problem 1. Double Counting

Let A, B, C be finite disjoint sets, with cardinalities $|A| = a$, $|B| = b$, and $|C| = c$. Let $S = \{x \cup y \cup z \mid x \subseteq A, y \subseteq B, z \subseteq C, |x| + |y| + |z| = n\}$. Now we can count the number of elements in S in two different ways.

First we can choose all possible combinations of x, y, z such that $|x| + |y| + |z| = n$, which gives us the following sum:

$$|S| = \sum_{i+j+k=n} \binom{a}{i}\binom{b}{j}\binom{c}{k}$$

Also we can choose n elements from the union of A, B, C, which gives us the following sum:

$$|S| = \binom{a+b+c}{n}$$

Thus we have the following identity:

$$\sum_{i+j+k=n} \binom{a}{i}\binom{b}{j}\binom{c}{k} = \binom{a+b+c}{n}$$

Problem 2. Pigeonhole Principle

We can express any number $x = q \cdot 2^k$ for some integers q and k with q odd.

Now in set $\{1, 2, 3, \ldots, 2n\}$ we have exactly n unique odd and even numbers. Now, since for every even number, its odd component satisfies $q < x$ and our series contains every odd number up to x, we don't need any new odd numbers to express even numbers. Thus, we can express every number from 1 to $2n$ using exactly n unique odd numbers.

Now, for the pigeonhole principle, each $n + 1$ elements of set S pigeon, and each unique odd number factor q needed to express that number is a pigeonhole, and each pigeon is assigned to a pigeonhole if they have same unique odd factor. Now, as shown above, we need exactly n unique odd numbers to express every number from 1 to $2n$.

Since $|S| = n + 1$, by the pigeonhole principle, there exist a, b such that they share the same unique odd factor; therefore a/b is a power of 2.

Problem 3. Dice Dice Dice Dice Dice

(a)

The set of outcomes of rolling 5 dice is $S = \{1, 2, 3, 4, 5, 6\}^5$ and the total number of outcomes is $|S| = 6^5$. And since the dice are fair it has a uniform probability distribution, so the probability of each outcome is $\frac{1}{6^5}$.

(b) $[A_{29}] = \{(6,6,6,6,6), (5,6,6,6,6), (6,5,6,6,6), (6,6,5,6,6), (6,6,6,5,6), (6,6,6,6,5)\}$

$$Pr[A_{29}] = \frac{6}{6^5}$$

(c) let outcome of our sample space be tuple $\omega = (d_1, d_2, d_3, d_4, d_5)$ and left function $f : [A_{18}] \to [A_{18}^c]$ be defined as

$$f(\omega) = (7 - d_1, 7 - d_2, 7 - d_3, 7 - d_4, 7 - d_5)$$

We know that for any $\omega \in A_{18}$, $S = \sum d_i$ and $18 \leq S \leq 30$. and now for $f(\omega), S' = \sum(7 - d_i) = 35 - \sum d_i = 35 - S$ which means

$$35 - 18 \geq 35 - S \geq 35 - 30$$

$$17 \geq S' \geq 5$$

Therefore resulting tuple $f(\omega) \in [A_{18}^c]$. Conversely, let $\omega \in A_{18}^c$, meaning its sum $5 \leq S \leq 17$. The sum of the mapped tuple $f(\omega)$ is $S' = 35 - S$. Since $S \leq 17$, it follows $35 - S \geq 35 - 17$, $S' \geq 18$. Therefore, $f(\omega) \in [A_{18}]$. Now showing f is a bijection by taking its own inverse

$$\begin{aligned} f(f(\omega)) &= 7 - (7 - d_1), 7 - (7 - d_2), \cdot, (7 - (7 - d_5)) \\ &= (d_1, d_2, d_3, d_4, d_5) \\ &= \omega \end{aligned}$$

hence f a bijaction between $[A_{18}]$ and it's complement.

Now $[A_{18}]$ and $[A_{18}^c]$ together cover the entire sample space since they are complements of each other; we have

$$Pr[A_{18}] + Pr[A_{18}^c] = 1$$

and also there is a bijection $Pr[A_{18}] = Pr[A_{18}^c]$ we have

$$2 \cdot Pr[A_{18}] = 1$$

$$Pr[A_{18}] = \frac{1}{2}$$

(d)

$$Pr[B_k] = \binom{5}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{5-k}$$

$$Pr[C_k] = \sum_{i=k}^{5} \binom{5}{i} \left(\frac{1}{6}\right)^i \left(\frac{5}{6}\right)^{5-i}$$

(e)

$$Pr[C_3|C_2] = \frac{Pr[C_3 \cap C_2]}{Pr[C_2]}$$

Since $C_3 \subseteq C_2$, $Pr[C_3 \cap C_2] = Pr[C_3]$ and using given probability from Part (d),

$$Pr[C_3|C_2] = \frac{276}{1526}$$

(f)

$$Pr[D_3|D_2] = \frac{Pr[D_3 \cap D_2]}{Pr[D_2]}$$

Since $D_3 \subseteq D_2$, $Pr[D_3 \cap D_2] = Pr[D_3]$, so:

$$Pr[D_3|D_2] = \frac{Pr[D_3]}{Pr[D_2]}$$

Now, $Pr[D_2] = 1 - Pr[D_2^c] = 1 - \frac{6!}{6^5} = \frac{7056}{7776}$.

For D_3, since at most one face can appear 3 or more times in 5 rolls, the events for each of the 6 faces to appear at least 3 times are mutually exclusive:

$$Pr[D_3] = 6 \cdot Pr[C_3] = 6 \cdot \frac{276}{7776} = \frac{1656}{7776}$$

Or via complement: $Pr[D_3] = 1 - Pr[D_3^c] = 1 - \frac{6! + 6\binom{5}{2}(5 \cdot 4 \cdot 3) + \binom{6}{2}\binom{5}{2}\binom{3}{2}(4)}{6^5} = 1 - \frac{6120}{7776} = \frac{1656}{7776}$

Therefore,

$$Pr[D_3|D_2] = \frac{1656/7776}{7056/7776} = \frac{1656}{7056} = \frac{23}{98}$$

(g)

- E_1: die 1 and 2 add up to 6
- E_2: die 2 and 3 add up to 6
- E_3: die 1 and 3 add up to 6

Now, by the inclusion-exclusion rule,

$[E_1 \cup E_2 \cup E_3] = |E_1| + |E_2| + |E_3| - |E_1 \cap E_2| - |E_2 \cap E_3| - |E_1 \cap E_3| + |E_1 \cap E_2 \cap E_3|$

There are a total of 5 possible pairs that sum up to 6, and the third number can be any of 6 on a die, so the total number of outcomes $= 5 * 6 = 30 = |E_1| = |E_2| = |E_3|$. Now for the set $\{E_1 \cap E_2\}$, here it only makes two pairs in one roll if die 1 and die 3 lead to the same result, and that result makes a pair with outcome of die 2, and since there are a total of 5 such outcomes to choose from for die 1 and die 2, $|E_1 \cap E_2| = 5$, and similarly $|E_2 \cap E_3| = |E_1 \cap E_3| = 5$ and $|E_1 \cap E_2 \cap E_3| = 1$ Because there is only one possible outcome $(3, 3, 3)$ where all three die pair up to 6.

$$Pr[E_1 \cup E_2 \cup E_3] = \frac{30 + 30 + 30 - 5 - 5 - 5 + 1}{6^3} = \frac{76}{216}$$

Problem 4. Unreliable Helpers

(a) Problem description in terms of E, T and L would be:

- $P(T|E) = 1 - 0.2 = 0.8$
- $P(L|E) = 1 - 0.25 = 0.75$
- $P(E) = 0.1$
- $P(T|E^c) = 0.2$
- $P(L|E^c) = 0.25$

(b)

$$\begin{aligned} Pr(E|T^c \cap L) &= \frac{Pr[E \cap T^c \cap L]}{Pr[T^c \cap L]} \\ &= \frac{0.1 * 0.2 * 0.75}{0.1 * 0.2 * 0.75 + 0.9 * 0.8 * 0.25} \\ &= 0.0769 \end{aligned}$$

(c) Intuition: They should not be equal because when LecBot say it has error then likeyhood of TeaBot respoding also saying it has error increases. Calculation:

$$Pr[T] = 0.1 * 0.8 + 0.9 * 0.2 = 0.26$$
$$Pr[L] = 0.1 * 0.75 + 0.9 * 0.25 = 0.3$$
$$Pr[T \cap L] = 0.1 * 0.8 * 0.75 + 0.9 * 0.2 * 0.25 = 0.105$$
$$Pr[T|L] = \frac{Pr[T \cap L]}{Pr[L]} = \frac{0.105}{0.3} = 0.35 \neq Pr[T]$$